

HW 2

Questions 1~4 carry a weight of **20** points while questions 5~6 carry a weight of **10** points. Furthermore, Questions 1~4 respectively have two sub-questions, a and b. TAs will evaluate using **5** sets of test data for each question (or sub-question), with each set of test data worth **2** points for questions 1~4, and **2** points for questions 5~6, respectively.

1. Reverse Words in a Sentence:

(a, 10%; b, 10%; total: 20%)

Write a program to reverse the order of words in a sentence.

- a. Use a loop
- b. Use **list slicing** only

Hint: for b, you may use **string.join**

Test data

Example 1:

input

hello world

output

world hello

Example 2:

input

python is fun

output

fun is python

2. Prime Number Generator:

(a, 10%; b, 10%; total: 20%)

Create a program that generates prime numbers within a specified range using a function, where the range is between $N1$ and $N2$, including $N1$ and $N2$, if $N1$ or $N2$ are prime numbers. Moreover, $N1$ and $N2$ are positive integers, $N1 < N2$, and $N1$ and $N2$ are separated by a space. Please output the prime numbers in one line in ascending order and separate them by commas. If there are no prime numbers, output 0.

- a. use for loop
- b. use while loop

Hint: You need to define a function in both sub-questions

Test data

Example 1:

input

2 33

output

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31

Example 2:

input

8 10

output

0

Example 3:

input

20 28

output

23

3. Remove Duplicates from List:

(a, 10%; b, 10%; total: 20%)

Write a program to remove duplicates from a list of numbers and output them in ascending order, where numbers are separated by commas.

- a. Use a loop
- b. Use a set

Test data

Example 1:

input

1, 2, 2, 3, 3, 3

output

1, 2, 3

Example 2:

input

5, 5, 5

output

5

Example 3:

input

5, 2, 1, 1, 3, 3, 6, 5, 2

output

1, 2, 3, 5, 6

4. Finding Common Elements in Lists:

(a, 10%; b, 10%; total: 20%)

Create a program that finds and prints the common elements between two lists, sorted in the following order: digits 0~9, uppercase letters A~Z, and lowercase letters a~z. If there are no common elements, please print "N/A". TA will give you two lines separately and there is no duplication in a line. Please use two input() to read input data. The first line is the first list and the second line is the second list. The elements in both lines are separated by commas.

- a. Loop is needed
- b. Define a function to do the comparison of two lists

Hint: the codes in b. should be different with those in a.

Test data

Example 1:

input

1, 2, 3, 4, 5

3, 4, 5, 6, 7

output

3, 4, 5

Example 2:

input

1, 2, 3, 4, 5

a, b, c, d, e, f

output

N/A

5. Convert Binary to Decimal:

Write a program to convert a binary number (given as a string) to a decimal number, where the converted decimal number is positive.

Hint: According to Wikipedia, a binary number is a number expressed in the base-2 numeral system or binary numeral system, a method for representing numbers that uses only two symbols for the natural numbers: typically "0" (zero) and "1" (one).

https://en.wikipedia.org/wiki/Binary_number

Test data

Example 1:

input

1010

output

10

Example 2:

input

1111

output

15

Example 3:

input

1011110100

output

756

6. Date Calculation:

Write a program that calculates the difference between two dates using the **datetime module**. The date format is YYYY-MM-DD, the unit of the difference is day, and two dates are in one line, separated by comma. If the input date format is incorrect or the date doesn't exist, please show "Invalid"

Hint 1: TAs may give you the invalid date format, such as 10-01-2026.

Hint 2: TAs may give you non-existence dates, such as 2026-09-31.

Hint 3: If the month or day is a single digit, automatically add a leading zero. Here is an example: 2023-1-2 → 2023-01-02.

Hint 4: The calculated result should always be non-negative. Remember to use the absolute value function, `abs()`.

Test data

Example 1:

input

1999-10-01, 2023-10-01

output

8766

Example 2:

input

2023-1-2, 2021-8-8

output

512

Example 3:

input

10-01-1999,2023-10-01

output

Invalid